Design and Implementation of a Multimodal Semantic Search Engine

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***Abstract*—In the last years, the amount of visual data available on the digital platforms has increased a lot, creating the necessity of retrieval systems that can go further into metadata and low-level visual similarity. The traditional image search approaches most times depend on the manual tags, filenames or captions, which are expensive to produce, inconsistent across users, and frequently incomplete. At the same time, classical computer vision methods based on handcrafted features can describe local visual patterns, but normally they are not able to capture the semantic meaning of an image. This paper presents the design and implementation of a multimodal semantic search engine which is capable of retrieving images from three different types of queries: the natural language text, spoken language by voice recognition, and input images. This system proposed combines different techniques that are CLIP for shared multimodal embeddings, Whisper for speech transcription, FAISS for efficient vector similarity search, and an A* diversification module to improve the quality of the final result gallery. The system is implemented as a full pipeline including an offline indexing stage, an online retrieval stage and a final interface web created with Gradio. The dataset used is Flickr8k and the experiments show a Recall@5 of 47% in a zero-shot setting over 100 random queries. In addition, the qualitative analysis shows that the integration of the heuristic search improves the diversity of the results and reduces the redundancy. This project demonstrates that combining deep learning and symbolic search can produce a more useful retrieval system than pure nearest-neighbor ranking alone.**

***Index Terms*—multimodal retrieval, semantic search, computer vision, natural language processing, speech recognition, vector databases, CLIP, FAISS, heuristic search**

I. Introduction

The growth of digital content has changed the way information is stored, shared, and consumed. Among all forms of digital data, images have a particularly important place because they are present in people's daily lives, and that means social media, online commerce, education, digital archives, surveillance systems, scientific repositories, and personal devices. Every day, a very large number of images are uploaded, stored and exchanged by users of the whole world. However, there is a problem, in many cases, the users do not know or remember the name of the file or the date they took the photograph, but instead, they remember the meaning of the photograph, for example, a kid playing basketball, a group of friends in front of a monument or a red car parked near a building.

This difference between machine representation and human interpretation is one of the central problems of multimedia retrieval. A computer stores an image looking at it as numerical values organized as pixels and channels, instead, a human looks at the image and interprets it as an object, an action, a context, or an event. In computer vision and information retrieval, this problem is commonly called the semantic gap (Smeulders et al., 2000). The semantic gap is the distance between low-level machine-readable information and high-level human semantic understanding. Solving this gap is essential if modern systems are expected to support natural and flexible search.

Traditional image retrieval systems are based on one of two strategies. The first strategy is metadata-based retrieval, in this one, the search depends on the tags, titles or captions that are associated with the image. This method can work sometimes if the images have complete and precise annotations but, on the other hand, this doesn't always happen, so it has clear disadvantages. Manual annotation is time-consuming, costly, and often inconsistent. Not everyone sees the image the same way, so different annotators may describe the image differently, and many image repositories do not contain enough textual information to support reliable search. The second strategy is content-based image retrieval. In this strategy the system uses visual descriptors like color, texture and shape to compare the images. This method is less dependent on the human work, it still does not solve the semantic problem, because visual resemblance does not necessarily imply conceptual similarity.

The limitations of traditional methods make multimodal semantic retrieval an attractive alternative. The idea is to connect images and language within a common representation space, instead of depending on tags or visual features created manually. With this system, we transform a text query like "a dog running on the grass" into a vector, and compare it with other image vectors. In the same way, an input image can be used as a query to retrieve other semantically related images and this creates a more natural and flexible way of interacting with large image collections.

This project proposes a multimodal semantic search engine designed around that idea. The system supports three modes of interaction. In the first one, the user writes a text describing the image he is searching for in natural language. In the second, the user speaks using a microphone, and the system transcribes the query (Radford et al., 2022). And in the third one, the user provides an image. These three modes are unified through a shared embedding space generated by a CLIP model (Radford

et al., 2021). The retrieval engine is built on a vector database using FAISS, and the final result set is refined through a symbolic A* search that reduces redundancy and promotes diversity.

The reasons that motivate this project are both practical and academic reasons. From a practical point of view, a multimodal search engine can improve accessibility and user experience. Voice interaction can be useful in areas like mobile environments and assistive applications, while image-to-image retrieval is useful in areas like browsing tasks and content exploration. If we look at the project from an academic perspective, this project is interesting because it combines important parts of artificial intelligence. The system we find at the end is a pipeline with which you can interact through a website.

For this project I selected the dataset Flickr8k, which has 8,000 images each one written by humans with five captions (Hodosh et al., 2013). This dataset is good for the project because it clearly connects what you see in an image, with the words used to describe it, and it allows both quantitative and qualitative evaluation of semantic retrieval.

These are the main contributions of the project. First, it has a full multimodal semantic retrieval pipeline that supports text, speech, and image queries. In second place, it combines the implementation of CLIP, Whisper, FAISS, and Gradio in a unified system. Third, in this work I introduced an A* post-processing stage to improve the system in the returned gallery with diversity. In last place, it evaluates the system quantitatively using Recall@5 and qualitatively through result inspection and failure analysis.

The remainder of this paper is organized in the following order. The next section is focused on the previous work and how this project fits in with other people's work. Then I will talk about the technical part, with all the background information you need to understand this proposal. The evaluation section comes after that, in which I will expand all the experimental parts and the results obtained. And finally the paper ends with the conclusion and the future work.

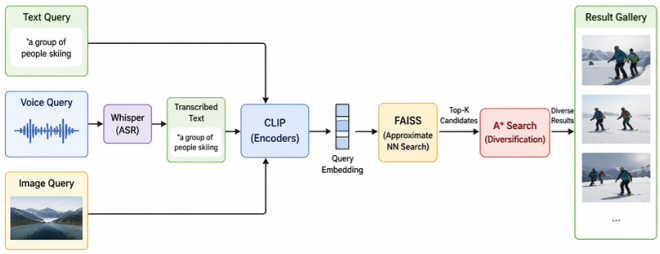

Fig. 1. Overview of the proposed multimodal semantic search system.

II. State of the Art

A. Classical content-based image retrieval

The first important generation of image retrieval systems was based on what is known as Content-Based Image Retrieval, or CBIR. In these systems the images are represented with low-level visual features such as color, texture, edges, shapes or patterns. The idea is that the images that look similar should be close in these types of details. This makes CBIR useful to find images with the same objects or visual browsing.

One of the most influential methods in this area is the Scale-Invariant Feature Transform, or SIFT. SIFT looks at local descriptors around key points of the image and describes what is around these points. This method is highly robust to changes in scale, rotation, and illumination (Lowe, 2004). This had a hard impact in computer vision because it provided a practical way to compare pictures even when they were changed in some way. However, SIFT and other similar methods can only represent patterns, not meanings. For example, this method can describe the appearance of a dog, but it can not tell you that it is actually a dog.

This semantic weakness is the principal limitation of early CBIR. Two images may appear similar because they share blue and white textures, even if one represents the sky and the other a painted wall. Therefore, CBIR works better than metadata-based systems in automatic retrieval, but it still did not bridge the semantic gap in a meaningful way.

B. Supervised deep learning approaches

Deep learning represented a step forward especially with convolutional neural networks. These convolutional neural networks or CNNs demonstrated that computers can learn from data, and their success in large-scale image classification changed the field completely. AlexNet was a milestone because it achieved a major performance improvement on ImageNet and showed the practical strength of deep convolutional architectures (Krizhevsky et al., 2012).

When the CNNs evolved they also became useful for retrieval, instead of using handcrafted features, they represent the image by the activations of hidden layers in a trained network. The representations of the images were more complex and abstract than color histograms or local descriptors and many times they contained the information about semantic categories. However, most of the CNNs continue being trained by humans with supervised setting with predefined class labels, and this created a limitation, the system was strong when the target concept matched the training categories but was not good enough for understanding natural language queries.

For example a CNN classifier may know the class "dog". It does not really understand a sentence like "a black dog jumping over a log in the forest". This is why the CNNs were a big improvement in this area but not the final answer to the problem of semantic retrieval.

C. Vision Transformers

The transformer architectures represent a more recent development in visual representation learning. What the vision transformers do is divide the image into patches, converting each patch into an embedding, and process the sequence through self-attention layers (Dosovitskiy et al., 2021). The difference between this and the CNNs is that the visual transformers do not rely on local convolutional kernels as

its primary inductive bias, it models global relations between patches more directly.

This is really important for semantic search, because a lot of searches need to understand the context of the relationships between things or what is happening in a scene. The transformer-based encoders can capture connections between different regions of the image in a more concrete way than classical CNN pipelines. This is one of the reasons why semantic search uses transformer-based backbones a lot in systems that deal with types of media like images and text, which is what we mean by multimodal systems.

D. Multimodal vision-language models

The strongest improvement in semantic retrieval is the multimodal representation learning. One of the most important examples of this is CLIP. It was trained on a large dataset of pictures and words and it learned to match what we see and what we read by comparing things that are not the same (Radford et al., 2021). What differs CLIP from other models is that CLIP makes it possible to compare any text query with an image representation, provided that both can be embedded into the same latent space, instead of classifying sort pictures into groups that are already defined.

This approach of CLIP is fundamental for zero-shot retrieval. The user can describe the image in natural language and the system will rank the images according to how similar to the description are, even if the concept explained by the user is not in the dataset. This makes CLIP be more flexible than older label-based approaches.

E. Multilingual and multimodal extensions

As I said, CLIP is powerful, but it is also limited by the language distribution of its training data, which is strongly biased toward English. This problem motivated future researchers into multilingual extensions, like mCLIP, which is based in CLIP with the purpose of extending this model to other languages that are not English. This is so important because image search should not depend only on English if we expect CLIP to expand to real-world global settings.

At the time multimodality is not just about images and text. Voice interaction is becoming incredibly important in user-facing systems, models such as Whisper, with robust speech recognition models, make it possible to connect what people say in spoken natural language with semantic retrieval pipelines (Radford et al., 2022). This means that a retrieval system can be more accessible to people letting them talk instead of typing.

F. Remaining gaps in the state of the art

Although multimodal models are strong, there are still some problems that need to be fixed. One important problem is the redundancy in the results, for example if the system looks for pictures that're really similar it might give us several pictures that are almost the same. This makes sense if we are just looking at how similar the pictures are. From the user's point of view it is not very helpful to see a lot of pictures that are almost identical. This is where the current project comes in, it adds a symbolic diversification stage on top of neural retrieval.

III. Preambles: Technical Background

In this section I will introduce the technical concepts that are important to understand the system I propose.

A. Flickr8k Dataset

The system of this project is built and evaluated in Flickr8k, this is a dataset with a bit more than 8,000 images and each one has five descriptions written by humans (Hodosh et al., 2013). This dataset is useful for this project because the captions can act as realistic text queries, while the images provide the retrieval target collection.

For the code, the images are read from the folder "data/Images/", and the evaluation script reads captions from "data/captions.txt". During the index part, we store image paths that are processed correctly along with their embeddings. With this we confirm that only the valid images are used during the retrieval and evaluation process.

B. Shared Embedding Spaces

The main idea of this system is to represent images and text in a common vector space (Radford et al., 2021). V_{img} is the visual embedding of an image and V_{txt} is the textual embedding of a query, so they should appear near one another in the latent space if they both describe the same idea.

Both text and image representations are encoded in 512-dimensional vectors in the CLIP model. Depending on the active query branch, the script specifically extracts "image_embeds" or "text_embeds" from the model outputs.

Fig. 2. CLIP multimodal embedding space. Text and image representations are projected into a shared vector space.

C. Cosine similarity and normalization

After embedding the system applies a L2 normalization to every vector, with the purpose of comparing the embeddings efficiently, this transforms each vector so its norm is 1. Then, cosine similarity can be calculated by computing the dot product. That is why the FAISS index being used in the project is IndexFlatIP, which means inner product. This is not only from a computational perspective but also provides stability when comparing vectors.

D. CLIP processing constraints in practice

One important practical aspect of the Hugging Face CLIP-Model is the possibility of being implemented in two different modalities of the input, there are text or images. For example, when using images as input, an empty string will be sent to the Hugging Face CLIPModel along with the image in order to create a representation for the image or, similarly, when sending text to the Hugging Face CLIPModel you can send it a 'dummy' black image in addition to the text. Both of these examples allow the user to complete the processor and forward pass for whichever model they choose.

E. Speech-to-text layer

For the voice interaction the model used is Whisper, which allows the user to record his voice in the Gradio interface. Then, Whisper transcribes the quote, so transform it to a text and this becomes the now retrieval query. For this project I used the base version of Whisper, which balances quality and practicality.

This architecture treats voice retrieval as a composition of two tasks: automatic speech recognition followed by semantic image retrieval.

F. Vector search

After the query is obtained, the method we use to find the nearest neighbors based on the embedding is FAISS (Johnson et al., 2017). Since the embedding matrix is pre-built and stored offline prior to the online processing phase, the only tasks performed during the online stage include loading, re-building, and searching the embedding index (matrix). The above process is quick from a computational standpoint and is consistent with the design of an interactive system.

G. Heuristic diversification

After the FAISS retrieval we use the algorithm A* (Russell & Norvig, 2010), we use it to improve the final output and reduce redundancy. What this algorithm does is to explore a pool of candidate images in subsets, using a cost function combining relevance and diversity. The Diversity penalty will be determined based on the average similarity between a newly selected candidate image and all of the images that were previously selected. The Relevance penalty will be based off of the original FAISS retrieved score. The weights assigned to each function is w1=0.7 and w2=0.3, allowing for a greater emphasis on semantic relevance, while providing a lesser penalty when redundant items are selected.

IV. Proposal and Approach

A. Overview of the proposed system

The proposed search engine separates the two processing phases: an Offline phase that sets up the searchable memory and an Online phase that processes user queries and returns results in real-time. This separation allows users to retrieve information quickly while processing queries using the Offline feature extraction process.

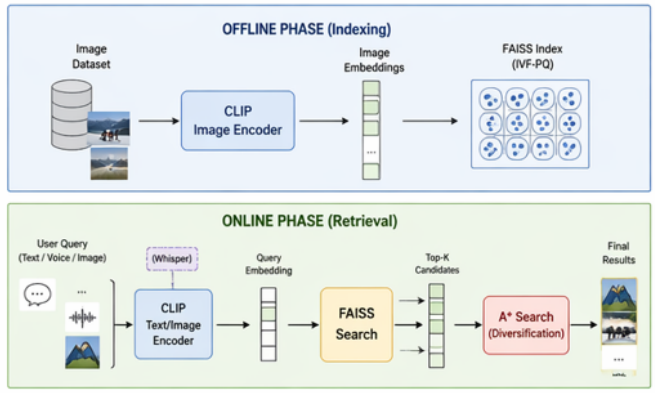

Fig. 3. System architecture showing the offline indexing phase and the online retrieval phase.

B. Offline phase: embedding extraction and storage

The offline phase starts with the dataset images, in which the script identifies all the ".jpg" files from the directory and then loads them one by one. In this process we convert the images to RGB format, this is needed to use the CLIP processor, because some user-provided or dataset images have incompatible channel formats such as grayscale or RGBA. By converting all the images to RGB we make sure that we have a consistent input.

The CLIP generates a 512-dimensional embedding for each of the images. Each of the embeddings is normalized using L2 Normalization for the purpose of performing a comparison using FAISS. Each normalized embedding vector is added to a list, and at the end the entire collection of normalized embedding vectors has been combined into a single matrix. The matrix containing all of the embedding vectors is saved to "image_embeddings.npy", and the paths to the images corresponding to those embeddings are saved to "image_paths.json".

C. Text query processing

When the user writes a text query, the system creates a dummy image to build the multimodal encoding needed by the CLIP processor. After that, the model introduces the text with the dummy image and it gets the tensor "text_embeds". This vector is normalized in a NumPy array that is compatible with FAISS. This design decision is significant because it demonstrates the practical application of the same multimodal model for text retrieval.

D. Speech query processing

For this project the voice pipeline is an extension of the text pipeline, because we use the Whisper model to transcribe the text so it converts the voice query into a text query. In practice, this means that automatic speech recognition and semantic image retrieval are combined to create speech retrieval, because it allows for multimodal interaction without creating a separate audio retrieval branch, this design is elegant.

E. Image-to-image retrieval

Apart from text and voice, the visual query also supports reverse image search. In this method the user uploads an image or a capture made with the webcam, and then, the system converts the input into a PIL image and forces it to RGB if it is needed, and extracts the "image_embeds" from the image CLIP visual branch. The last step is to compare against the dataset the resulting vector normalized. This functionality is especially useful because many times the user wants to find images similar to one they already have instead of searching for images through a description, or in cases in which the user does not know how to describe the image he is looking for but he has a similar photograph.

F. Initial retrieval with FAISS

After obtaining the embeddings of the query, the system uses the FAISS model to find the most similar images to the text, voice or image the user inputs. This model generates an output of five images that are returned directly. In the advanced A* script and in the application, the system will first get a pool of 20 candidates to search through. This larger pool of candidates will then be processed by a diversification module to narrow them down to the most relevant results.

G. A* diversification module

The A* module is the fundamental part of this project. This algorithm chooses a subset of five images from the candidate pool of the FAISS model and uses a cost function that incorporates both diversity and semantic relevance to assess candidate combinations. So the top five most similar items are not the only ones that the algorithm considers and this incorporation substantially improves the result of FAISS.

The FAISS similarity score is used to calculate relevance. The function "calculate_diversity_penalty" calculates diversity by comparing a new candidate to the state's previously chosen images. The penalty increases if the new candidate closely resembles those pictures. After that, the algorithm searches for a group of candidates that collectively minimize the cost using A*.

This implementation makes the candidate indices grow in increasing order, so the algorithm focuses on combinations instead of permutations, and this makes the method computationally manageable and prevents needless search explosion.

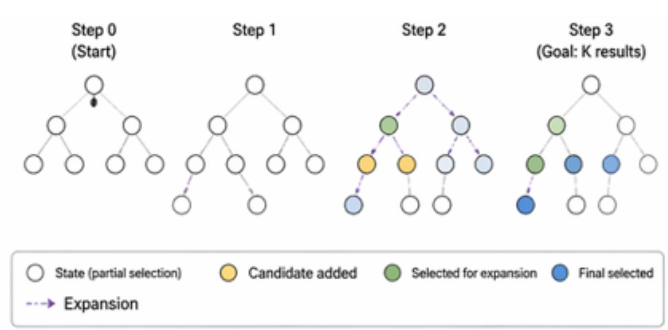

Fig. 4. Illustration of the A* search process used to select a diverse subset of images.

H. Interactive application

The final part of this project is the implementation of the full system in the Gradio interface. The Gradio website presents a simple interface in which, in first place, are three boxes with the three types of inputs, one to put the text query, other with the button to record your voice with the query you want to search for and another with the option to upload an image from your own gallery. Then, the interface has a status field and on the right part the result gallery with the five images selected by the model. This interface transforms the technical pipeline into a complete application that, in a few seconds, can be tested interactively.

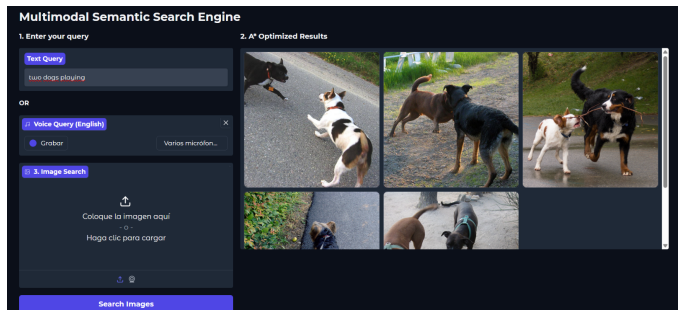

Fig. 5. User interface of the multimodal semantic search system implemented with Gradio.

V. Evaluation and Validation

A. Experimental Setup

For the evaluation of this project I used the Flickr8k dataset. The first thing I do for the experimental part is to store image paths and the offline embedding matrix. Then, the evaluation script reads the captions file and reconstructs the FAISS index. Using a fixed seed of 42, it selects a random sample of 100 valid images, filters out any image not in the processed index, and maps image names to their available captions. This configuration is crucial to make the experiment repeatable.

B. Evaluation Metric

The metric used for the evaluation is Recall@5 (Manning et al., 2008). What this metric does is select one image and generate a query to try to target that image, and check if it appears in the top five retrieved results, if the target image is in the top five it counts as a hit. The system does that for all the images and the final Recall@5 score is the number of hits divided by the total number of evaluated queries.

This is the appropriate metric to evaluate this project because the user typically only sees a small gallery of results, so in practical search systems, usually it is more crucial that the pertinent result appears early than at a lower rank.

C. Evaluation procedure

There is a clear protocol for the evaluation of the model. One of the five available captions is selected randomly for every image in the dataset, and that caption becomes the textual query. That query goes through a process of being

encoded with CLIP and then searched against the FAISS index, and after that, the output is examined to determine whether the target image appears in the top five results.

An important aspect of this method is that only the query is considered correct if it hits the target image in the output, so the query is considered unsuccessful if the system returns multiple images that are semantically similar but does not include the original image. That makes the metric conservative.

D. Quantitative Results

The implemented evaluation script reports the following final result:

$$Recall@5 = 47.00\% \tag{1}$$

TABLE I
QUANTITATIVE EVALUATION RESULTS

Metric	**Value**
Dataset	Flickr8k
Embedding Model	CLIP ViT-B/32
Query Type	Text queries (captions)
Number of Queries	100
Top-K	5
Recall@5	47.00%

For a zero-shot retrieval system, this is a strong result. Although the model was not specifically optimized for Flickr8k retrieval, it is nearly always able to place the right image in the top five. This result suggests a significant level of semantic alignment between image and text, given the thousands of images in the search space and the visual similarity of many scenes in Flickr8k.

E. Interpretation of the quantitative results

We will now interpret that 47%. This dataset contains many images that are quite similar between them, like children playing, dogs running pictures of a mountain, and we need to keep in mind that we have a dataset of more than 8,000 images. So many times the system retrieves images that are semantically correct but are not the target image we are searching for, so in those cases, that result does not count as correct.

For this reason, Recall@5 is a good metric for this word but it only gives a partial view for the evaluation, and we have to complete it with the qualitative analysis.

F. Qualitative analysis

For this analysis we make manual tests, and the system performs a good behavior. Relevant images were typically returned by queries like "a dog playing with a ball" or "a group of people skiing". The main problem I had with the FAISS results was redundancy, because many times the selection of images for the output extremely similar between each others, so I fixed this problem with the A* algorithm, after applying A* the final subset had far more variety between the output images, but still preserving the main semantic concept.

TABLE II
COMPARISON BETWEEN FAISS AND FAISS + A*

Method	Relevance	Diversity
FAISS only	High	Low
FAISS + A*	High	High

This is a significant finding because diversity frequently affects user satisfaction. Although practically weak, a gallery with five nearly identical images might be mathematically correct. A gallery that presents slightly different perspectives on the same idea is frequently more beneficial.

G. Error analysis

Ambiguity is one of the main errors seen in this system. First of all, long descriptions in natural language generates more specific retrieval results that short queries with poor descriptions, that generates worse results. Another source of errors is with Whisper, when it misidentifies the audio input, that leads to problems with the speech transcription, so the retrieval module gets the incorrect text and consequently looks for the incorrect idea.

The manual setting of the weighting parameters in the A* diversification step is a third drawback. While the selected values function fairly well, they might not be ideal for all types of queries.

H. Comparison with simpler alternatives

Although the project lacks a comprehensive numerical benchmark against supervised CNN baselines or SIFT-based retrieval, the conceptual comparison is evident. The current system supports free-text and voice queries and reasons at a semantic level in contrast to CBIR. It allows zero-shot retrieval using natural language, in contrast to fixed-label CNN systems. It enhances the final result gallery by adding a symbolic diversification layer in comparison to pure CLIP plus FAISS retrieval.

VI. CONCLUSIONS

This paper presents a multimodal semantic search engine that can retrieve images from text, speech, and visual queries. The system integrates FAISS for effective nearest-neighbor retrieval, Whisper for automatic speech transcription, A* for final result diversification, and CLIP as a shared vision-language encoder.

With this work I demonstrate that a multimodal design improves semantic retrieval. The system lets users express their query more naturally rather than limiting them to metadata or fixed labels. Additionally, it demonstrates that retrieval quality is not solely dependent on semantic similarity. Additionally, diversity is crucial, and by eliminating redundancy, the symbolic search stage enhances the result gallery's utility.

The implementation is complete and practical. It consists of an interactive Gradio application, an offline extraction pipeline, an online search pipeline, and a diversification module. In a zero-shot setting over 100 random queries, the evaluation on Flickr8k yielded a Recall@5 of 47%, confirming that the

system captures a significant relationship between images and language. Simultaneously, qualitative analysis demonstrates that the A* module clearly adds value beyond a straightforward nearest-neighbor search.

The project's primary finding is that a retrieval system can be made more powerful by combining deep learning and traditional heuristic search than by using either strategy alone. While the symbolic algorithm enhances the returned gallery's structure, the neural model offers semantic alignment.

VII. Future Work

In the last part I am going to expose several realistic directions for future work.

The first approach is multilingual retrieval. The current CLIP-based text retrieval stage is still heavily focused on English, despite Whisper's multilingual transcription capabilities. Performance for Spanish and other languages could be enhanced by substituting the existing encoder with a multilingual variant like mCLIP (Chen et al., 2023).

Scalability is the second direction. For Flickr8k, exact FAISS search is suitable; however, much larger datasets would necessitate approximate nearest-neighbor techniques like IVF or HNSW.

Learned diversification is the third approach. Currently, fixed weights govern how diversity and relevance are balanced. This trade-off could be discovered by a more robust system in the future through user feedback or interaction logs.

The fourth direction is domain adaptation. Fine-tuning or parameter-efficient adaptation could greatly improve retrieval in specialized fields like medicine, engineering, law, or cultural heritage.

A more comprehensive assessment is the fifth and last direction. To better capture semantic usefulness, future research could incorporate metrics like MRR, NDCG, or human evaluation protocols in addition to Recall@5.

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